SignSaathi: M2 Progress Report

Accessible AI for All | Scenario 3: City Signage | Team: Inference Lab | 11 October 2026

PROBLEM REFINED AFTER USER RESEARCH	CURRENT PROGRESS
Navigation apps can bring a user near a destination, but blind and low-vision users still need help identifying the exact shop, entrance, bus board, platform or signal. SignSaathi focuses on this final, high-friction part of the journey.	Field visit and contextual interviews, workflow mapping, competitor review, primary persona, user journey, safety rules and system architecture are complete. Bhashini, camera/OCR and orchestration integration are in progress.

1. User research completed

The team visited **Asha Kuteer Home for the Blind**, a disability services and support organisation in Begumpet, Hyderabad. We spoke with a partially blind participant, a fully blind participant and a volunteer, using a contextual interview and task walkthrough to understand how people currently use phones, screen readers and navigation tools outside the home. Participants are kept anonymous in this report.

What we observed	What it means for the product
Participants use TalkBack on Android and tools such as NVDA or JAWS on computers. Voice navigation is already familiar.	The interface must be voice-first, fully labelled for TalkBack and usable without visual confirmation.
Google Maps helps with the route, but users still ask people to find the exact bus, shop, entrance or boarding point.	The product should solve the last-50-metres gap by combining location context with camera-based sign verification.
Metro travel becomes easier where tactile paths, Braille signs and staff assistance exist. Buses and street crossings create more uncertainty.	Start with shop boards, bus boards, platforms and exits. Treat pedestrian-signal information as assistance, never a guarantee that crossing is safe.
Blur, camera tilt and fast-moving objects reduce accuracy. A fully blind user cannot know whether the sign is inside the frame.	Add spoken alignment cues, frame-quality checks and automatic capture only after the phone becomes stable.
Users preferred a useful detailed description and wanted to interrupt speech to ask a follow-up question.	Use layered, interruptible audio: answer first, then direction, supporting detail and an option to hear more.
Traffic noise affects microphone input and makes voice output difficult to hear.	Confirm critical speech, combine voice with vibration and provide recovery when speech recognition is uncertain.

2. Primary persona

PROFILE	GOALS	BEHAVIOURS	PAIN POINTS
Blind or low-vision Android user who travels with a white cane and depends on audio rather than the screen.	- travel with less dependence on strangers - identify the exact destination - receive information in a familiar language	- uses TalkBack and voice input - uses maps or ride-hailing - holds the cane and phone together	- uncertain final location - camera framing - noise and unreliable recognition - unsafe confident answers

3. Conditions supported and how the system infers them

Condition	How SignSaathi adapts	Signals used to infer it
Visual (primary)	Voice-first flow, spoken camera-alignment cues, clock-direction and distance-range answers, vibration confirmations	User starts with voice, makes no screen taps, or moves through the page with screen-reader swipe patterns
Limited hand control	Hands-free capture once the phone is stable, every control available by voice, large touch targets, no timed gestures	Repeated missed or accidental taps, failed long-presses, several abandoned capture attempts
Ageing	Slower speech, large high-contrast text, one instruction at a time, one-tap "Explain simpler" and "Repeat"	Frequent repeat requests, pinch-zoom, long pauses between steps

The system proposes a profile in the user's language and asks for confirmation (for example, "Shall I speak more slowly?"). The user can change it at any time, and inference signals are kept on the device for the current session only.

4. Accessible output: raw vs SignSaathi (design target)

Plain AI assistant reply	SignSaathi reply for the same photo and question
"The image shows a storefront with a blue signboard that reads Sri Sai Medical and General Stores. Below it there is a phone number and timings from 9 AM to 10 PM. Two people are standing near the entrance and a scooter is parked on the left..."	"This is **Sri Sai Medical Store**. The entrance is at about 1 o'clock, a few steps ahead. It is open until 10 PM. Say *more* for details." **Low confidence:** "I can read *Sri Sai Med*, but the rest is cut off. Move the phone slightly right and hold still."

Illustrative example of the output rules; the side-by-side comparison will be measured on real sign photos during testing (Section 7).

Solution approach and implementation progress

One complete journey is prioritised: reach the destination area, verify the exact place and ask a follow-up question in the user's language.

5. Proposed user journey and features

1 ASK	2 ROUTE	3 ALIGN	4 VERIFY	5 GUIDE	6 RECOVER
Speak a destination or question in Hindi, Telugu or English.	Location guidance brings the user near the selected place.	Voice and vibration help position the camera; stable frame auto-captures.	OCR reads exact text; vision and nearby-place data establish context.	Answer gives identity, clock direction, distance range and useful detail.	Low confidence triggers a retake, confirmation or human-help request.

CORE FEATURES	ACCESSIBLE OUTPUT	SAFETY AND RECOVERY
- shop and sign identification - bus-board and platform reading - last-50-metres destination verification - camera alignment and auto-capture	- Hindi, Telugu and English - interrupt, repeat and slower controls - clock-direction and distance cues - TalkBack labels plus vibration	- confidence shown in words - OCR, vision and location agreement - retake instead of guessing - consent-based human help fallback

6. Technical implementation

Layer	Implementation	Responsibility
Frontend	Android Chrome PWA; camera, microphone and location permissions; TalkBack; frontend state machine; frame-quality Web Worker	Manage interaction, permission recovery, camera alignment, interruption and accessible feedback.
Backend	Python FastAPI; explicit asynchronous orchestration; REST for media; WebSocket for progress and cancellation	Coordinate services, run OCR/vision/location work in parallel and record stage latency.
Language	Bhashini ASR, NMT, TTS and OCR for English, Hindi and Telugu	Convert voice to text, translate the request and return the verified answer as speech.
Reasoning	Vision-capable analysis plus Llama API for intent and response composition	Combine evidence into a structured answer; the model does not independently approve safety.
Validation	Deterministic Python safety gate with confidence, agreement and stale-location checks	Return verified answer, retake request or consent-based human-in-the-loop webhook.

7. Experiments run and progress on testing

Activity	Status	Outcome / next measurement
Field visit, contextual interviews and travel-task walkthrough (Asha Kuteer)	**Completed**	Established the last-50-metres problem, output preferences, camera-framing barrier, noise issue and trust requirements.
Existing-tool and workflow review	**Completed**	Mapped TalkBack, Maps, scene-description and accessible-navigation tools; defined the missing link between route and exact-place verification.
Informal static camera-description trial	**Completed**	Static description worked in a controlled case; moving objects and blurred images remain outside the core MVP.
Failure-mode and safety review	**Completed**	Defined retake, confirmation, timeout, disagreement and human-help branches. No silent guessing is allowed.
Bhashini speech pipeline	**In progress**	Measure ASR recognition, translation quality, TTS clarity and end-to-end latency for all three languages.
Camera, OCR and TalkBack testing	**In progress**	Test blur, tilt, low light, traffic noise, permission denial, network loss and eyes-free task completion.
Raw vs accessible comparison	**Planned**	Same sign photos and questions to a plain assistant and to SignSaathi; compare answer length, time to the key fact and clarity rating.

8. M2 code-push scope

frontend/	permission flow, TalkBack-ready interface, voice controls, camera alignment, condition profiles and accessible output
backend/	FastAPI endpoints, Bhashini/vision/location adapters, orchestration, validation gate and HITL webhook
tests/	sample signs, noisy audio, blur/tilt cases, network failure cases and a small observed-output log
README	setup, environment variables, run steps, API/model declarations, known limitations and demo flow